SYSTEM DESIGN

# Load Balancing Visual Guide

How traffic gets distributed across servers for scale, speed, and survival.

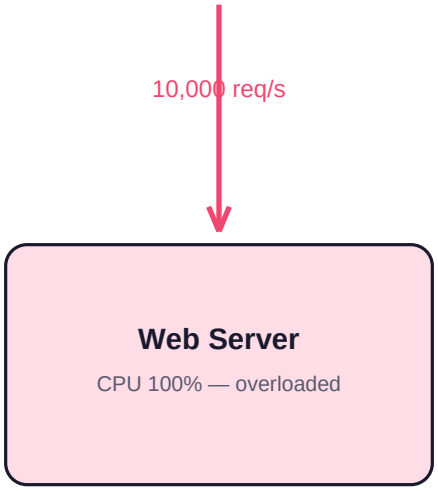
**ANIMATED WALKTHROUGH**

Step through pages to play the animation

STEP 1/6

# The Problem: One Server Can't Handle It All

A single server is a single point of failure — and a bottleneck.

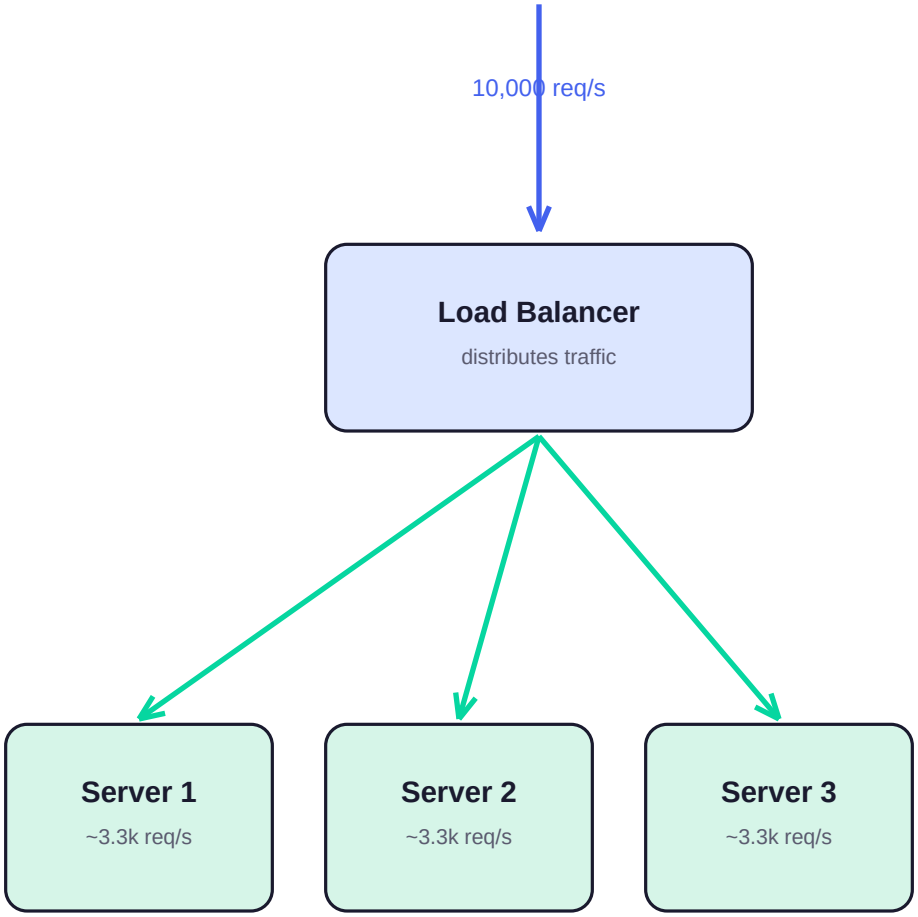



One server receiving all traffic: slow responses, crashes under load, zero fault tolerance.

STEP 2/6

# Insert a Load Balancer

A reverse proxy in front that spreads requests across N servers.




Each backend now handles a fraction. Add servers to scale horizontally.

STEP 3/6

# Distribution Algorithms

How the balancer picks the next server.

**Round Robin**

1 → 2 → 3 → 1 → 2 → 3 …

**Least Connections**

Send to the server with fewest active conns

**IP Hash**

Same client → same server (sticky sessions)

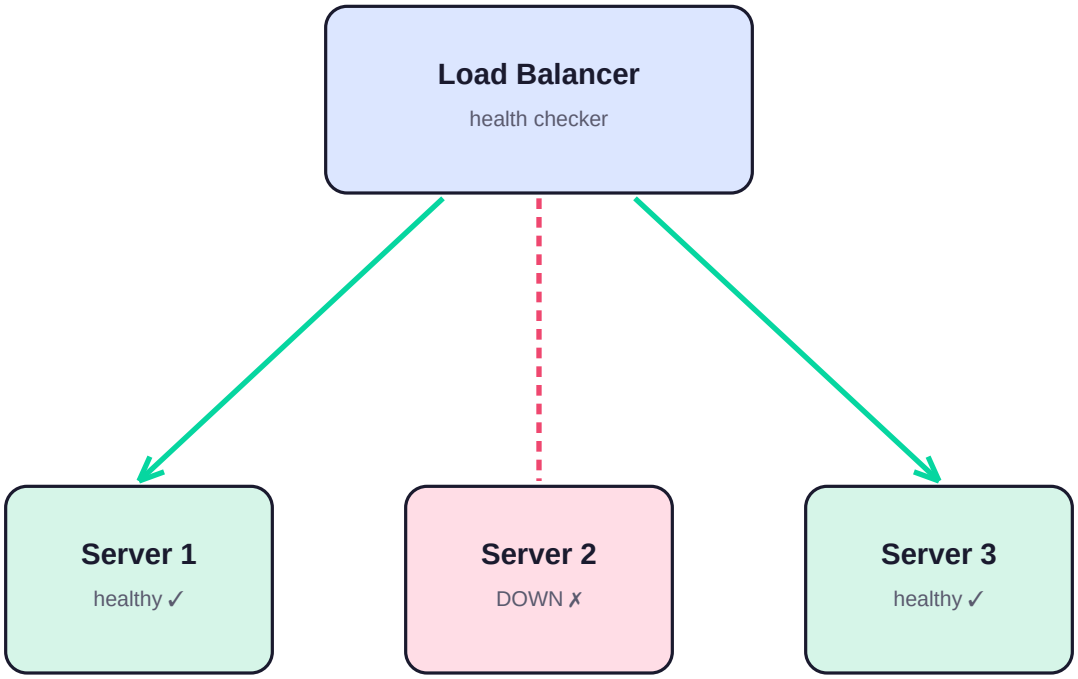
**Weighted**

Bigger servers get proportionally more traffic

STEP 4/6

# Health Checks: Route Around Failure

The balancer probes backends and stops sending traffic to dead ones.




Failed probes remove Server 2 from rotation automatically. Traffic keeps flowing.

STEP 5/6

# L4 vs L7 Load Balancing

## Layer 4

TCP/UDP — fast, dumb

- Routes by IP + port
- No request inspection
- Very high throughput
- Example: NLB, HAProxy TCP

## Layer 7

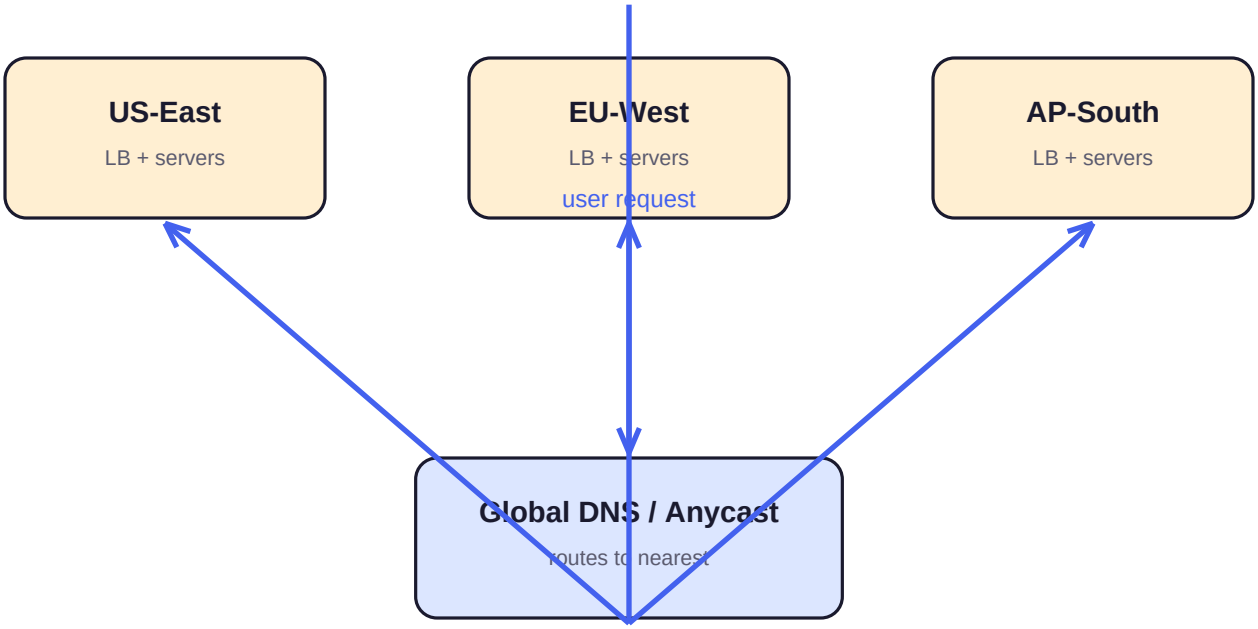
HTTP — smart routing

- Routes by URL, header, cookie
- TLS termination, rewrites
- Content-based rules
- Example: ALB, nginx, Envoy

STEP 6/6

# Going Global: DNS + Anycast

Multiple regions, each with its own balancer tier.




Global load balancing sends each user to the closest healthy region — lower latency, disaster resilience.

STEP ✓/✓

# Key Takeaways

## Load Balancing — Remember

- A load balancer spreads traffic so no single server is a bottleneck or single point of failure.
- Algorithms: round robin (simple), least connections (dynamic), IP hash (sticky), weighted (heterogeneous).
- Health checks automatically route around dead backends.
- L4 is fast packet routing; L7 understands HTTP and routes on content.
- Global LB (DNS/anycast) directs users to the nearest healthy region.

**Keep exploring**

Pair this guide with the matching video walkthroughs and hands-on labs at interstitiumlabs.dev